

# A BIQUATERNIONIC SOURCE-CLOSURE DERIVATION OF THE BORN RULE AT THE $\delta$ -KERNEL LIMIT AND CLOSURE ORDER $\alpha = 2$ OF THE CAUSAL ACCUMULATION LAW

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**ABSTRACT.** We give a self-contained algebraic derivation of the projective Born rule from the Causal Accumulation Law (CAL) in the Markovian ( $\delta$ -kernel) limit at the closure order  $\alpha = 2$ . The carrier is the biquaternion algebra  $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ . The source is not a bare quaternion and not a single unembedded complex scalar: each quaternionic coefficient is supplied by its own holomorphic two-channel current  $Z_\mu = A_\mu e^{-H_\mu/\lambda_\mu} + B_\mu e^{iS_\mu/\hbar_{\text{eff},\mu}}$ , and the physical current is the embedded source-current velocity  $\mathcal{J}(\mathbf{Z}) = q_0(Z_0) + q_1(Z_1)\mathbf{i} + q_2(Z_2)\mathbf{j} + q_3(Z_3)\mathbf{k} \in \mathbb{C} \otimes \mathbb{H}$ . This  $\mathcal{J}(\mathbf{Z})$  is tangent/current data, not a state vector; the projector is the normalized readout of its null direction. We therefore present two equivalent Born routes: the ordinary state-vector/projector route after representing  $\mathbb{C} \otimes \mathbb{H}$  as  $M_2(\mathbb{C})$ , and the intrinsic source-current rotor route after the  $\alpha = 2$  closure map has produced the same Hermitian idempotent. The raw holomorphic coefficients do not by themselves imply  $2J = |J|^2$ ; the equality appears at the normalized stationary image of the embedded current. At  $\alpha = 2$  this image is the Hermitian null-idempotent class  $J_\pm(\mathbf{n}) = 1 \pm i(n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k})$ ,  $\|\mathbf{n}\| = 1$ , satisfying  $J_\pm^\dagger = J_\pm$ ,  $J_\pm^2 = 2J_\pm$ , and  $N(J_\pm) = 0$ . Hence  $P_\pm = J_\pm/2$  are complete orthogonal projectors. The internal trace readout  $\tau_{\mathcal{B}}(q) = 2\text{ReSc}(q)$  gives  $\tau_{\mathcal{B}}(P_\pm(\mathbf{m})P_+(\mathbf{n})) = (1 \pm \mathbf{m} \cdot \mathbf{n})/2$ , i.e.  $\cos^2(\Theta/2)$  and  $\sin^2(\Theta/2)$ . The signature statement is the split carrier  $(4,4)$  with Hermitian projection to  $(1,3)$ , not a  $(2,2) \rightarrow (1,3)$  reduction. No Gleason-type premise, external partition integral, or auxiliary Hilbert-space postulate is invoked.

## CONTENTS

|                                                                   |    |
|-------------------------------------------------------------------|----|
| 1. Scope and claim                                                | 1  |
| 2. Definitions                                                    | 2  |
| 3. Structural relations                                           | 4  |
| 4. Algebraic lemmas                                               | 5  |
| 5. The closure at $\alpha = 2$ and the $\delta$ -kernel selection | 6  |
| 6. Main theorem                                                   | 7  |
| 7. Remarks on scope                                               | 9  |
| Appendix A. Notation                                              | 10 |
| References                                                        | 11 |

## 1. SCOPE AND CLAIM

This paper isolates one cell of the parent framework's ( $\alpha$ , kernel) table [1, §IX]: the corner  $\alpha = 2$  with memory kernel  $k(\tau; x, x') = \delta(\tau - \tau_0)$ . It is therefore a companion to the broader field theory, but not a dependent appendix to it. Because this limiting cell closes algebraically, the sector has its own internal burden of proof: the relevant source embedding, conjugations, closure

relation, projectors, and Born weights are extracted here and derived in full. The parent paper supplies provenance and the larger landscape; this paper supplies the standalone limiting-case derivation. The claim is the following.

*Given the parent accumulation relation (Relation 3.1), the channel-wise holomorphic source current embedded in  $\mathbb{C} \otimes \mathbb{H}$  (Relation 3.2), and the order-two source closure derived from the product/conjugation structure (Proposition 5.2), the normalized  $\alpha = 2$  source image is a Hermitian null idempotent. Its halves are complete orthogonal projectors, and the internal scalar trace forces the probabilities  $p_{\pm} = (1 \pm \mathbf{m} \cdot \mathbf{n})/2$  without a measurement postulate. The same probability can be read in two equivalent presentations: one-sided state-vector action in a matrix representation, or the intrinsic two-sided rotor/sandwich action on a reference idempotent. The dual-conjugate structure of the biquaternions and the integration order  $\alpha = 2$  do the selecting.*

The derivation is algebraic and exact; it does not pass through Gleason's theorem [2], does not assume a probability measure on Hilbert subspaces, and does not impose an external normalization. The measurement layer is derived as the projector algebra of the normalized null direction of the embedded source-current velocity; the rotor/sandwich presentation is therefore primary for CAL, while the state-vector formula is the standard matrix representation of the same projector. The Lorentzian signature selection  $(4, 4) \rightarrow (1, 3)$  and the Born weight are shown to be two readings of one Hermitian projection (Corollary 6.2). Everything used here is stated as a definition, an inherited structural relation, or a derived proposition; the only imports from the parent manuscript are the constraint labels (C6, C8, C13–C15, C17–C19 of [1, §I]) cited for provenance, and the fixed-point uniqueness result cited in Remark 7.3.

## 2. DEFINITIONS

**Definition 2.1** (Causal past and memory kernel). For an event  $x$  on a Lorentzian manifold,  $J^-(x)$  denotes its past light cone. A *memory kernel* is a normalized, causal, non-negative weight  $k(\tau; x, x')$  on proper-time separation  $\tau$  with support in  $J^-(x)$ . The  $\delta$ -kernel (Markovian) limit is  $k(\tau; x, x') = \delta(\tau - \tau_0)$ , which selects the unique past point  $x'$  at proper-time displacement  $\tau_0$ .

**Definition 2.2** (Fractional order and phase bookkeeping). The *fractional order*  $\alpha \in [0, 4)$  is the single dial of the accumulation relation, weighing present against memory. The symbol  $\theta_\alpha$  is phase bookkeeping for the product sector associated with order  $\alpha$ , not an independent axiom. It is

$$(1) \quad \theta_\alpha = \frac{\alpha\pi}{2},$$

the phase label that also appears in the fractional product representation: acting on a phase eigenfunction  $e^{i\omega s}$ , the Caputo fractional integral of order  $\alpha$  multiplies by  $(i\omega)^{-\alpha} = \omega^{-\alpha} e^{-i\alpha\pi/2}$  [3, 4]. At the integer corners, the corresponding phase signs are  $e^{i\theta_\alpha} = 1, i, -1, -i$  for  $\alpha = 0, 1, 2, 3$ . In the present derivation the value  $\theta_\alpha = \pi$  at  $\alpha = 2$  is explained by the intrinsic bar/dagger product closure in Proposition 5.2; it is not assumed as an extra rule. The symbol  $\Theta$  is reserved below for the relative analyzer/projector angle in the Born rule.

**Definition 2.3** (Biquaternion algebra). The *biquaternions* are the tensor product  $\mathbb{C} \otimes \mathbb{H}$  over  $\mathbb{R}$ , with quaternion units  $i_q = \mathbf{i}, \mathbf{j}, \mathbf{k}$  and a privileged *central* imaginary  $i$  from the  $\mathbb{C}$  factor, commuting with all elements. A general element is

$$(2) \quad q = q_0 + q_1 \mathbf{i} + q_2 \mathbf{j} + q_3 \mathbf{k}, \quad q_n \in \mathbb{C}.$$

As an associative algebra,  $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ .

**Definition 2.4** (The two conjugations). On  $\mathbb{C} \otimes \mathbb{H}$  define:

- (i) the *quaternionic conjugation* (bar),  $\bar{q} = q_0 - q_1\mathbf{i} - q_2\mathbf{j} - q_3\mathbf{k}$ , which negates the quaternion units and leaves the complex coefficients untouched, so  $\bar{i} = +i$ ;
- (ii) the *dagger conjugation*,  $q^\dagger = \bar{q}_0 - \bar{q}_1\mathbf{i} - \bar{q}_2\mathbf{j} - \bar{q}_3\mathbf{k}$ , which additionally complex-conjugates the coefficients, so  $i^\dagger = -i$ .

The associated quadratic objects are the *reduced norm*  $N(q) = q\bar{q} = q_0^2 + q_1^2 + q_2^2 + q_3^2$  and the *Hermitian scalar self-product*  $\langle qq^\dagger \rangle_0 = |q_0|^2 + |q_1|^2 + |q_2|^2 + |q_3|^2$ , where  $\langle \cdot \rangle_0$  denotes the scalar (grade-0) part.

**Definition 2.5** (Hermitian and anti-Hermitian projections).  $\mathcal{H}(q) = \frac{1}{2}(q + q^\dagger)$  and  $\mathcal{A}(q) = \frac{1}{2}(q - q^\dagger)$ . Under  $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ ,  $\mathcal{H}$  selects the Hermitian slice  $X = X^\dagger$ , a real four-dimensional subspace on which the reduced norm acts as the determinant  $\det X = t^2 - x^2 - y^2 - z^2$ . Thus the correct carrier statement is split signature (4, 4) with Hermitian projection to (1, 3); no (2, 2) antecedent is used.

**Definition 2.6** (Channel-wise source embedding). For each biquaternionic channel  $\mu = 0, 1, 2, 3$ , let

$$(3) \quad Z_\mu = A_\mu \exp\left[-\frac{H_\mu}{\lambda_\mu}\right] + B_\mu \exp\left[\frac{iS_\mu}{\hbar_{\text{eff},\mu}}\right],$$

with  $i$  the central imaginary of the  $\mathbb{C}$  factor. The source-current velocity entering the algebra is the embedded biquaternion

$$(4) \quad \mathcal{J}(\mathbf{Z}) = q_0(Z_0) + q_1(Z_1)\mathbf{i} + q_2(Z_2)\mathbf{j} + q_3(Z_3)\mathbf{k} \in \mathbb{C} \otimes \mathbb{H},$$

where the  $q_\mu$  are the channel coordinates supplied by the corresponding holomorphic source currents. This object is a velocity/current vector in the source fiber, not a state vector. The identity  $2J = |J|^2$  is not asserted for arbitrary raw coefficients  $q_\mu(Z_\mu)$ ; it is the normalized null-idempotent closure of the embedded current direction.

**Definition 2.7** (Null channels and trace readout). For a unit vector  $\mathbf{n} = (n_1, n_2, n_3) \in \mathbb{R}^3$  define

$$(5) \quad e_{\mathbf{n}} = n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k}, \quad E_{\mathbf{n}} = i e_{\mathbf{n}}.$$

The normalized closed source currents and their projectors are

$$(6) \quad J_\pm(\mathbf{n}) = 1 \pm E_{\mathbf{n}}, \quad P_\pm(\mathbf{n}) = \frac{1}{2}J_\pm(\mathbf{n}) = \frac{1}{2}(1 \pm E_{\mathbf{n}}).$$

The scalar readout used for probabilities is the internal matrix trace of the biquaternion presentation,

$$(7) \quad \tau_{\mathcal{B}}(q) := 2 \operatorname{Re} \operatorname{Sc}(q) = 2 \operatorname{Re} \langle q \rangle_0.$$

On Hermitian products this is real and agrees with the ordinary matrix trace under  $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ .

**Definition 2.8** (Currents, projector states, and transition amplitudes). Present and past CAL inputs are realized embedded source-current velocities  $v = \mathcal{J}(x)$  and  $v_0 = \mathcal{J}(x')$ , valued in the source-current fiber carried by  $\mathbb{C} \otimes \mathbb{H}$ . They are tangent/current data, not state vectors. At the normalized  $\alpha = 2$  closure, their null directions define projector states  $\rho = P_\pm(\mathbf{n})$  of Definition 2.7. When a central scalar transition amplitude is used,  $W = \langle \psi | \psi_0 \rangle$  denotes the overlap of the induced projector states  $\psi, \psi_0$ ; its squared modulus is the trace-product probability derived in Theorem 6.1, not an independently postulated Hilbert-space rule for the raw velocity vector.

**Definition 2.9** (Rotor closure of a current direction). Let  $P_0 = P_+(\mathbf{k})$  be a fixed reference idempotent. In the compact projective sector, a normalized rotor  $R \in \mathbb{C} \otimes \mathbb{H}$  satisfies  $RR^\dagger = 1$  and acts on Hermitian objects by the sandwich map

$$(8) \quad X \longmapsto R X R^\dagger.$$

For a unit quaternionic generator  $\mathbf{u}^2 = -1$ , a physical rotation angle  $\varphi$  is represented by the rotor half-angle

$$(9) \quad R(\varphi) = \exp(-\mathbf{u}\varphi/2) = \cos \frac{\varphi}{2} - \mathbf{u} \sin \frac{\varphi}{2}.$$

The half-angle is not an added trigonometric rule: the sandwich action applies  $R$  and  $R^\dagger$ , so two half-angles produce the net angle  $\varphi$ . The source-current velocity route is therefore the closure map

$$(10) \quad \mathcal{J}(\mathbf{Z}) \longmapsto R(\mathbf{Z}) \longmapsto \rho_R(\mathbf{Z}) = R(\mathbf{Z})P_0R(\mathbf{Z})^\dagger.$$

When the closed current has null direction  $\mathbf{n}$ , this gives  $\rho_R = P_\pm(\mathbf{n})$ . Thus a generic velocity vector is not assigned a Born probability by itself; it yields a Born probability only after the  $\alpha = 2$  closure map has turned its normalized direction into a Hermitian idempotent. In the two-plane reference-overlap case, choose  $R(\theta_\alpha)$  so that the resulting idempotent has analyzer angle  $\Theta = \theta_\alpha = \alpha\pi/2$  relative to  $P_0$ . Then

$$(11) \quad \tau_B\left(P_0R(\theta_\alpha)P_0R(\theta_\alpha)^\dagger\right) = \cos^2 \frac{\theta_\alpha}{2} = \cos^2 \frac{\alpha\pi}{4}.$$

This is a reference-channel overlap, while the general measurement probability in Theorem 6.1 depends on the analyzer angle  $\Theta$  between arbitrary prepared and measured projectors.

### 3. STRUCTURAL RELATIONS

**Relation 3.1** (Parent accumulation relation; C8 of [1]). The parent CAL relation, quoted here rather than re-axiomatized, writes the observable at  $x$  as the sum of the immediate causal impulse and the memory-weighted, transported past:

$$(12) \quad T(x) = \alpha J(x) - (1 - \alpha) \int_{J^-(x)} k(\tau; x, x') \mathcal{P}(x, x') J(x') d^4x',$$

where  $\mathcal{P}(x, x')$  collects the bitensor transport, parallel propagators, and fiber holonomy, and reduces to the identity in the flat-field limit.

**Relation 3.2** (Two-channel source current and embedding; C17–C18 of [1]). The source current has two channel factors in every biquaternionic coefficient: a real thermal (cost) factor and a unitary phase (action) factor, as in (3). The current that enters the parent accumulation relation is the embedded source-current velocity (4), not a state vector. Centrality of  $i$  is forced: the action phase is a global object and must not depend on a preferred quaternionic direction. The minimal algebra simultaneously supporting a commuting phase unit and three anticommuting rotational generators is  $\mathbb{C} \otimes \mathbb{H}$  (Definition 2.3); quaternions alone lack the commuting unit, and complex numbers alone lack the rotational generators.

**Relation 3.3** (Amplitude-overlap notation). In the  $\delta$ -kernel transition, the scalar amplitude notation  $W = \langle \psi | \psi_0 \rangle$  denotes the central overlap induced by the same dagger form on the projector states determined by the embedded source-current velocities. It is a shorthand for the trace-product closure of Theorem 6.1, not an independent Hilbert-space postulate.

These relations are the inherited CAL structure and notation used in the proof. The channel angle is not listed here as an axiom: it is the phase bookkeeping of Definition 2.2, and its order-two value is derived from the conjugate product and norm closure in Proposition 5.2. Reality, positivity, orthogonality, and normalization are outputs of that closure, not assumptions about an apparatus.

#### 4. ALGEBRAIC LEMMAS

**Lemma 4.1** (Complementarity of the two closures). *On  $\mathbb{C} \otimes \mathbb{H}$ :*

- (a) *the reduced norm is multiplicative,  $N(qr) = N(q)N(r)$ , complex-valued, and indefinite; under  $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$  it is the determinant, vanishing exactly on the zero divisors (the null cone) [5];*
- (b) *the Hermitian scalar self-product  $\langle qq^\dagger \rangle_0$  is positive definite, vanishing only at  $q = 0$ , and is not multiplicative;*
- (c) *the two are complementary on the null cone: the explicit zero divisor  $q = 1 + i\mathbf{i}$  (central imaginary times quaternion unit; the product  $h = i\mathbf{i}$  is the split unit,  $h^2 = i^2\mathbf{i}^2 = (-1)(-1) = +1$ ) has  $N(q) = 0$  yet  $\langle qq^\dagger \rangle_0 = 2 > 0$ .*

*Proof.* (a) is the statement that  $N$  coincides with  $\det$  under the matrix isomorphism, and  $\det(QR) = \det Q \det R$ ; the vanishing locus of  $\det$  is the zero-divisor variety. (b)  $\langle qq^\dagger \rangle_0 = \sum_n |q_n|^2$  is a sum of complex moduli, hence non-negative and zero only when every  $q_n = 0$ ; a direct two-element product shows failure of multiplicativity. (c) For  $q = 1 + i\mathbf{i}$ , the complex coefficients are  $q_0 = 1$ ,  $q_1 = i$ , so  $N(q) = q_0^2 + q_1^2 = 1 + i^2 = 0$ : the element sits on the null cone,  $\det = 0$  (verified numerically for this element in the companion suite [1, §V E]), while  $\langle qq^\dagger \rangle_0 = |q_0|^2 + |q_1|^2 = 1 + 1 = 2$ . The bar carries the indefinite metric channel; the dagger carries the positive event weight; one algebra carries both.  $\square$

**Lemma 4.2** (Signature selection). *The Hermitian projection  $\mathcal{H}(q) = \frac{1}{2}(q + q^\dagger)$  maps the split-signature  $(4, 4)$  carrier onto a real four-dimensional slice of Lorentzian signature  $(1, 3)$ , annihilating the complementary anti-Hermitian summand of signature  $(3, 1)$ : the carrier splits as  $(4, 4) = (1, 3) \oplus (3, 1)$  [1, Part VII]. On the surviving slice the reduced norm acts as  $\det X = t^2 - x^2 - y^2 - z^2$ . A basis of the slice is  $\{1, i\mathbf{i}, i\mathbf{j}, i\mathbf{k}\}$ : the scalar direction is timelike, while the spatial coefficients enter the reduced norm as  $(ix)^2 = -x^2$ ,  $(iy)^2 = -y^2$ , and  $(iz)^2 = -z^2$ .*

*Proof.* The Hermitian slice of  $M_2(\mathbb{C})$  is the real span of  $\{1, \sigma_x, \sigma_y, \sigma_z\}$ ; the determinant of  $X = t1 + x\sigma_x + y\sigma_y + z\sigma_z$  is  $t^2 - x^2 - y^2 - z^2$ , signature  $(1, 3)$ . The stated biquaternionic basis is the preimage of the Pauli basis under the isomorphism. The kernel of  $\mathcal{H}$  is the anti-Hermitian summand, the real span of  $\{i1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ , on which the real part of  $N$  has signature  $(3, 1)$ ;  $\mathbb{C} \otimes \mathbb{H}$  is eight real dimensions and  $(1, 3) + (3, 1) = (4, 4)$  accounts for all of them. This is the corrected signature chain; a  $(2, 2)$  intermediate carrier would have the wrong real dimension.  $\square$

**Lemma 4.3** (Null-idempotent source projectors). *For every unit  $\mathbf{n} \in \mathbb{R}^3$ , the elements of Definition 2.7 satisfy*

$$(13) \quad J_\pm^\dagger = J_\pm, \quad J_\pm^2 = 2J_\pm, \quad J_\pm J_\pm^\dagger = 2J_\pm, \quad N(J_\pm) = 0,$$

and

$$(14) \quad P_\pm^2 = P_\pm, \quad P_\pm^\dagger = P_\pm, \quad P_+ P_- = 0, \quad P_+ + P_- = 1.$$

*Proof.* Since  $e_{\mathbf{n}}^2 = -\|\mathbf{n}\|^2 = -1$  and the central imaginary satisfies  $i^2 = -1$ , one has  $E_{\mathbf{n}}^2 = (ie_{\mathbf{n}})^2 = +1$ . Also  $E_{\mathbf{n}}^\dagger = E_{\mathbf{n}}$ , because dagger complex-conjugates the coefficient  $i$  and reverses the quaternionic vector sign, producing two minus signs. Therefore  $J_\pm^\dagger = J_\pm$  and  $J_\pm^2 = (1 \pm E_{\mathbf{n}})^2 =$

$2(1 \pm E_{\mathbf{n}}) = 2J_{\pm}$ . The bar conjugate is  $\overline{J_{\pm}} = 1 \mp E_{\mathbf{n}}$ , so  $N(J_{\pm}) = J_{\pm}\overline{J_{\pm}} = 1 - E_{\mathbf{n}}^2 = 0$ . Dividing  $J_{\pm}$  by 2 gives (14).  $\square$

**Lemma 4.4** (Uniqueness of the conjugation-invariant quadratic form). *Up to positive scale, the Hermitian self-product  $W^{\dagger}W$ , read through its scalar part  $\langle W^{\dagger}W \rangle_0$ , is the unique real, non-negative, conjugation-invariant quadratic form on  $\mathbb{C} \otimes \mathbb{H}$ . In particular the plain square  $W^2$  is inadmissible: the quaternionic part is non-commutative and the complex part is non-real, so  $W^2$  does not land in the observable (real scalar) sector, and the bar-channel product  $\langle W\bar{W} \rangle_0$  retains an imaginary interference part because  $\bar{i} = +i$ .*

*Proof.* A quadratic form on  $\mathbb{C} \otimes \mathbb{H}$  built from one conjugation is either the bar form or the dagger form (Definition 2.4), and the two involutions divide the required properties between them: quaternion reversal  $\sigma$  (the bar) supplies the grade/metric structure but yields the indefinite complex-valued  $N(W)$  (Lemma 4.1(a)), real reality failing because  $\bar{i} = +i$ ; complex-coefficient conjugation  $\kappa$  alone supplies complex conjugacy but not a quaternionic positive form. Only their composite  $\dagger = \sigma \circ \kappa$  supplies both at once, giving  $\langle WW^{\dagger} \rangle_0 = \sum_n |W_n|^2$ , real and non-negative (Lemma 4.1(b)); invariance is immediate since  $\langle (W^{\dagger})^{\dagger}W^{\dagger} \rangle_0 = \langle WW^{\dagger} \rangle_0 = \langle W^{\dagger}W \rangle_0$  on the scalar sector.  $W^2$  is excluded as stated. This is a statement about the algebra alone; that the dagger form is moreover the *operative* closure, rather than one admissible form among several, is fixed by the integration order in Proposition 5.2, where  $\alpha = 2$  rotates it onto the real axis. No further independent quadratic combination of  $W$  with its conjugates produces a real non-negative scalar not proportional to the dagger form.  $\square$

**Lemma 4.5** (Interference cancellation at the dagger). *Let  $W = g + i t_p$  with  $g, t_p \in \mathbb{R}$  and  $i$  the central imaginary. Then*

$$(15) \quad W^{\dagger}W = (g - i t_p)(g + i t_p) = g^2 + t_p^2 = |W|^2,$$

*the would-be interference  $\pm i g t_p$  cancelling identically because  $i^{\dagger} = -i$ . In the full algebra, writing  $W = G + S$  with  $G^{\dagger} = G$ ,  $S^{\dagger} = -S$ , the scalar closure is*

$$(16) \quad \text{tr}(W^{\dagger}W) = \|G\|^2 + \|S\|^2 + \text{tr}[G, S],$$

*and the entire interference is the commutator  $[G, S] = GS - SG$ , which is traceless and contributes nothing to the scalar observable, identically and not as an approximation.*

*Proof.* Equation (15) is direct multiplication using  $i^{\dagger} = -i$  and centrality of  $i$ . For (16):  $W^{\dagger}W = (G - S)(G + S) = G^2 - S^2 + [G, S]$  on the split  $G^{\dagger} = G$ ,  $S^{\dagger} = -S$ ; the squares contribute  $\|G\|^2 + \|S\|^2$  to the trace (the anti-Hermitian  $S$  contributing  $-S^2 = S^{\dagger}S \geq 0$ ), and  $\text{tr}[G, S] = 0$  by cyclicity of the trace.  $\square$

## 5. THE CLOSURE AT $\alpha = 2$ AND THE $\delta$ -KERNEL SELECTION

**Proposition 5.1** (Sign flip and doubled present). *At  $\alpha = 2$ , the memory prefactor in (12) is  $-(1 - \alpha) = +1$ : present and past are cooperative, with the present weighted double. With the  $\delta$ -kernel of Definition 2.1 and flat transport, the accumulation update is*

$$(17) \quad T|_{\alpha=2, \delta} = 2J(x) + J(x') = 2v + v_0.$$

*Proof.* Substitute  $\alpha = 2$  and  $k = \delta(\tau - \tau_0)$  into (12); the integral collapses to the single past point  $x'$  with coefficient  $-(1 - 2) = +1$ , and  $\mathcal{P} \rightarrow \mathbf{1}$ .  $\square$

**Proposition 5.2** (Emergent order-two source closure). *At  $\alpha = 2$ , the parent relation gives the immediate source term  $\alpha J = 2J$ . On the embedded, stationary, normalized source image,*

the intrinsic bar/dagger product structure forces the same doubled object to be the Hermitian quadratic norm:

$$(18) \quad \alpha J_{\pm}|_{\alpha=2} = 2J_{\pm} = J_{\pm}J_{\pm}^{\dagger} = |J_{\pm}|_{+}^2.$$

The phase label  $\theta_{\alpha}$  is therefore not an independent angle postulate. At order two it records the antipodal conjugate branch of the product closure:  $\theta_{\alpha} = \pi$  and  $e^{i\theta_{\alpha}} = -1$ .

*Proof.* The first equality is the parent relation's present coefficient evaluated at  $\alpha = 2$ . The rest is forced by the intrinsic conjugates and norms. Put  $E_{\mathbf{n}} = ie_{\mathbf{n}}$  with  $e_{\mathbf{n}}^2 = -1$ . Dagger changes both signs,  $i \mapsto -i$  and  $e_{\mathbf{n}} \mapsto -e_{\mathbf{n}}$ , so  $E_{\mathbf{n}}^{\dagger} = E_{\mathbf{n}}$ ; bar changes only the quaternionic sign, so  $\overline{E_{\mathbf{n}}} = -E_{\mathbf{n}}$ . Since  $i^2 = e_{\mathbf{n}}^2 = -1$ , their product satisfies  $E_{\mathbf{n}}^2 = +1$ .

Now write a normalized two-channel source image as  $J_{\lambda} = 1 + \lambda E_{\mathbf{n}}$  with  $|\lambda| = 1$ . The Hermitian quadratic product is

$$(19) \quad J_{\lambda}J_{\lambda}^{\dagger} = 2 + (\lambda + \bar{\lambda})E_{\mathbf{n}},$$

whereas the doubled linear current is  $2J_{\lambda} = 2 + 2\lambda E_{\mathbf{n}}$ . Equality forces  $\lambda = \bar{\lambda}$ , hence  $\lambda = \pm 1$ . Thus the conjugate product does not allow an arbitrary phase at closure; it collapses the source image to the two antipodal real branches  $J_{\pm} = 1 \pm E_{\mathbf{n}}$ . These branches are separated by the phase  $-1 = e^{i\pi}$ , so the nontrivial conjugate/memory branch has  $\theta_{\alpha} = \pi$ . This is exactly the same sign reached by the order-two phase bookkeeping  $e^{i\alpha\pi/2}$  at  $\alpha = 2$ , but here it follows from the product, dagger, bar, and norm closure themselves. Lemma 4.3 then gives  $J_{\pm}J_{\pm}^{\dagger} = 2J_{\pm}$  and  $N(J_{\pm}) = 0$ .  $\square$

**Proposition 5.3** (Equivalence of state-vector and rotor presentations). *Let  $P_0 = P_{+}(\mathbf{k})$  and let  $R \in \mathbb{C} \otimes \mathbb{H}$  be a normalized rotor with  $RR^{\dagger} = 1$  whose sandwich action sends the reference idempotent to  $P_{+}(\mathbf{n})$ :*

$$(20) \quad \rho_R = RP_0R^{\dagger} = P_{+}(\mathbf{n}).$$

*Under any matrix representation  $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ , choose a unit column  $V_0$  satisfying  $V_0V_0^{\dagger} = P_0$  and set  $V = RV_0$ . Then*

$$(21) \quad VV^{\dagger} = RP_0R^{\dagger} = \rho_R.$$

*Consequently, for every projective effect  $P_a$ ,*

$$(22) \quad \text{Tr}(P_aVV^{\dagger}) = \tau_{\mathcal{B}}(P_aRP_0R^{\dagger}).$$

*The one-sided state-vector calculation and the two-sided rotor calculation are therefore the same probability calculation after projection to a density/idempotent.*

*Proof.* Associativity gives  $VV^{\dagger} = R(V_0V_0^{\dagger})R^{\dagger} = RP_0R^{\dagger}$ . The trace identity follows because  $\tau_{\mathcal{B}}$  is the internal matrix trace of the biquaternion presentation on Hermitian products (Definition 2.7).  $\square$

## 6. MAIN THEOREM

**Theorem 6.1** (Projective Born rule at  $\alpha = 2$ ,  $\delta$ -kernel). *With  $k(\tau; x, x') = \delta(\tau - \tau_0)$  and  $\alpha = 2$ , let  $\rho = P_{+}(\mathbf{n})$  be the projector state induced by the normalized present source-current velocity. A two-outcome projective measurement along unit direction  $\mathbf{m}$  has effects  $P_{+}(\mathbf{m})$  and  $P_{-}(\mathbf{m})$ . The probabilities forced by the biquaternionic trace readout are*

$$(23) \quad \mathbb{P}(\pm | \mathbf{n}, \mathbf{m}) = \tau_{\mathcal{B}}(P_{\pm}(\mathbf{m})\rho) = \frac{1 \pm \mathbf{m} \cdot \mathbf{n}}{2}.$$

If  $\Theta$  is the measurement angle between  $\mathbf{m}$  and  $\mathbf{n}$ , then

$$(24) \quad \mathbb{P}(+ | \mathbf{n}, \mathbf{m}) = \cos^2 \frac{\Theta}{2}, \quad \mathbb{P}(- | \mathbf{n}, \mathbf{m}) = \sin^2 \frac{\Theta}{2}.$$

Equivalently, the same probability has both one-sided state-vector and two-sided rotor presentations,

$$(25) \quad \mathbb{P}(+ | \mathbf{n}, \mathbf{m}) = \text{Tr}(P_+(\mathbf{m})V_{\mathbf{n}}V_{\mathbf{n}}^\dagger) = \tau_{\mathcal{B}}(P_+(\mathbf{m})R_{\mathbf{n}}P_0R_{\mathbf{n}}^\dagger),$$

where  $V_{\mathbf{n}}V_{\mathbf{n}}^\dagger = P_+(\mathbf{n})$  and  $R_{\mathbf{n}}P_0R_{\mathbf{n}}^\dagger = P_+(\mathbf{n})$ . In central scalar transition-amplitude notation,

$$(26) \quad P(\psi_0 \rightarrow \psi) = \frac{W^\dagger W}{\langle \psi | \psi \rangle \langle \psi_0 | \psi_0 \rangle} = \frac{|\langle \psi | \psi_0 \rangle|^2}{\langle \psi | \psi \rangle \langle \psi_0 | \psi_0 \rangle}.$$

*Proof.* The  $\delta$ -kernel picks the unique past point  $x'$  and the accumulation update is  $T = 2v + v_0$  by Proposition 5.1. The probability-bearing object is not the raw update  $T$ ; it is the normalized closed source image  $J_\pm = 1 \pm E_{\mathbf{n}}$  selected by the  $\alpha = 2$  channel closure (Proposition 5.2). By Lemma 4.3,  $P_\pm = J_\pm/2$  are Hermitian, idempotent, orthogonal, and complete. Thus they already carry the projective measurement algebra. Proposition 5.3 gives the two equivalent presentations of this algebra: the standard one-sided state-vector representation  $V_{\mathbf{n}}V_{\mathbf{n}}^\dagger = P_+(\mathbf{n})$ , and the intrinsic rotor representation  $R_{\mathbf{n}}P_0R_{\mathbf{n}}^\dagger = P_+(\mathbf{n})$ . The latter is the same sandwich action used for Hermitian spacetime vectors and velocity directions, but here it acts on the reference idempotent rather than on a raw vector.

It remains to compute the trace readout. Quaternion multiplication gives

$$(27) \quad e_{\mathbf{m}}e_{\mathbf{n}} = -\mathbf{m} \cdot \mathbf{n} + e_{\mathbf{m} \times \mathbf{n}},$$

and therefore, because  $i^2 = -1$ ,

$$(28) \quad E_{\mathbf{m}}E_{\mathbf{n}} = (ie_{\mathbf{m}})(ie_{\mathbf{n}}) = \mathbf{m} \cdot \mathbf{n} - e_{\mathbf{m} \times \mathbf{n}}.$$

The scalar part of (28) is  $\mathbf{m} \cdot \mathbf{n}$ . Hence

$$(29) \quad \tau_{\mathcal{B}}(P_+(\mathbf{m})P_+(\mathbf{n})) = 2 \text{Sc} \left[ \frac{1}{4}(1 + E_{\mathbf{m}})(1 + E_{\mathbf{n}}) \right]$$

$$(30) \quad = \frac{1 + \mathbf{m} \cdot \mathbf{n}}{2}.$$

Replacing  $\mathbf{m}$  by  $-\mathbf{m}$  gives the minus outcome. The two probabilities add to one because  $P_+(\mathbf{m}) + P_-(\mathbf{m}) = 1$  and  $\tau_{\mathcal{B}}(\rho) = 1$ ; they lie in  $[0, 1]$  because  $-1 \leq \mathbf{m} \cdot \mathbf{n} \leq 1$ . The half-angle form follows from  $\mathbf{m} \cdot \mathbf{n} = \cos \Theta$ . This  $\Theta$  is the analyzer/projector angle, not the CAL channel angle  $\theta_\alpha = \alpha\pi/2$ . In the special reference-overlap case where the rotor moves  $P_0$  by the CAL channel angle, (11) gives  $\cos^2(\alpha\pi/4)$ ; at  $\alpha = 2$  this is the orthogonal reference-channel overlap selected by closure, not a claim that all measurement probabilities vanish. Finally, the scalar amplitude form (26) is the central-coordinate presentation of the same dagger trace closure, with interference cancelled as in Lemma 4.5.  $\square$

**Corollary 6.2** (Born  $\equiv$  Lorentzian projection). *The Born weight and the signature selector are one algebraic readout. Read as a probability rule, the Hermitian projectors and the trace  $\tau_{\mathcal{B}}$  yield non-negative normalized event weights (Theorem 6.1); read as a signature selector, the same dagger/Hermitian structure yields the realized (1,3) Lorentzian slice of the split (4,4) carrier (Lemma 4.2). These are two readings of one move, not two coincident results.*

**Corollary 6.3** (The closure is the closed amplitude, not the accumulation current). *For normalized velocities in the dagger form, the raw self-overlap of the accumulation update (17) is*

$$(31) \quad \langle T | T \rangle = 5 + 4 \text{Re} \langle v | v_0 \rangle,$$

which carries only the linear interference term and not the squared modulus. The Born probability is the norm of the closed amplitude  $W$ , equivalently the trace product of the closed source projectors, not the norm of the raw accumulation current  $T$ ; conflating the two is the error the closure structure of Proposition 5.2 forecloses.

*Proof.*  $\langle 2v + v_0 | 2v + v_0 \rangle = 4 + 1 + 4 \operatorname{Re} \langle v | v_0 \rangle$  for unit-normalized velocities in the dagger form.  $\square$

**Corollary 6.4** (Well-posedness on the light cone). *The Born weight remains well defined on the null cone: the reduced norm degenerates exactly there, as the causal structure requires, while the Hermitian closure stays positive on the same locus (Lemma 4.1(c)). The signed metric rides the bar channel; the positive event weight rides the dagger channel.*

**Corollary 6.5** (Differential of the Hermitian closure). *The closure is stable under variation with the same factor of two the parent manuscript's scalar identity carries. For any variation  $dW$ ,*

$$(32) \quad d \langle WW^\dagger \rangle_0 = 2 \operatorname{Re} \langle (dW)W^\dagger \rangle_0,$$

the two conjugate contributions  $(dW)W^\dagger$  and  $W(dW)^\dagger = ((dW)W^\dagger)^\dagger$  being daggers of one another and therefore summing, on the scalar part, to twice the real part of either. This is the biquaternionic lift of the gradient identity  $\nabla A = \frac{C}{2} \nabla C$  at  $A = C^2/4$  [1, Thm. IX.6]: differentiating the real scalar square produces the factor of two by the product rule, and the dual-conjugate structure is what makes the two members conjugate rather than independent. The factor is thus algebraic, one contribution from each conjugate orientation, and not a normalization inserted by hand.

*Proof.* Since  $\dagger$  is an anti-involution (Definition 2.4),  $((dW)W^\dagger)^\dagger = W(dW)^\dagger$ , so the product-rule expansion  $d(WW^\dagger) = (dW)W^\dagger + W(dW)^\dagger$  is a sum of a term and its dagger. On the scalar part  $\langle q^\dagger \rangle_0 = \overline{\langle q \rangle_0}$ , hence  $\langle (dW)W^\dagger \rangle_0 + \langle W(dW)^\dagger \rangle_0 = \langle (dW)W^\dagger \rangle_0 + \overline{\langle (dW)W^\dagger \rangle_0} = 2 \operatorname{Re} \langle (dW)W^\dagger \rangle_0$ .  $\square$

## 7. REMARKS ON SCOPE

*Remark 7.1* (Why Gleason is not needed). Gleason's theorem [2] assumes a Hilbert space and additivity of a measure over orthogonal decompositions, then deduces a trace rule. Here the order is reversed: the embedded source current closes to explicit orthogonal idempotents,  $P_+ + P_- = 1$  is computed inside  $\mathbb{C} \otimes \mathbb{H}$ , and the trace readout  $\tau_{\mathcal{B}}(P\rho)$  is forced by the same dagger form that supplies positivity. Gleason becomes a parallel result one can cite rather than a prerequisite one must invoke [1, §VI B]. The statement is not that  $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$  has no matrix or spinor representation; it is that no auxiliary Hilbert-space probability postulate is used to obtain the rule.

*Remark 7.2* (Which presentation is primary). The state-vector formula is kept because it is the standard textbook presentation and follows immediately in any  $M_2(\mathbb{C})$  representation. For CAL, however, the rotor/sandwich presentation is structurally primary:  $\mathcal{J}(\mathbf{Z})$  is source-current velocity data, and the same two-sided map  $X \mapsto RXR^\dagger$  acts on Hermitian spacetime vectors, velocity directions, and projector idempotents. The Born rule is not assigned to an arbitrary velocity vector; it is read only after the normalized  $\alpha = 2$  closure maps the current direction to a Hermitian idempotent. In that sense the rotor route is the better intrinsic derivation, while the state-vector route is the familiar representation of the same projector algebra.

*Remark 7.3* (Uniqueness and stability of the measure). The present paper establishes that the  $\delta$ -kernel,  $\alpha = 2$  closure is the Born transition probability and that no other real, non-negative

closure is admissible on  $\mathbb{C} \otimes \mathbb{H}$  at this order. That the Born measure is additionally the unique, globally attracting fixed point of the local-access map  $T : p_\Psi \mapsto \mathcal{N} \cdot W^\dagger W[p_\Psi]$ , with explicit contraction constant  $\kappa' \leq (1 - \nu_{\min})/2 < \frac{1}{2}$  via the Dobrushin coefficient of the Rényi averaging kernel, is Theorem VI.1 of [1] and is cited, not re-proved. The two results are complementary readings at different kernel widths: the present theorem is the single-transition ( $\delta$ ) seed; the fixed point governs the full non-local structure.

*Remark 7.4* (Position within the  $\alpha = 2$  family). Born is the lower end of a continuous family parameterized by kernel width at the closure corner:  $\delta$ -kernel gives the transition probability derived here; the full-width retarded kernel gives the Feynman path sum, coinciding with the standard path integral under the channel-split lemma and measure-reduction condition of [1, Thm. IX.4], a conditional result outside the present scope. The scalar identity uniting Lagrangian, Legendre transform, and Born as one quadratic closure at  $\alpha = 2$  is [1, Thm. IX.6].

*Remark 7.5* (Deviation handles). The derivation gives the Born rule at the product-closure point and at  $\alpha = 2$ . Two independent predictive handles exist for deviations: closure failure (the rank-2 pipeline rails carrying inconsistent invariants,  $\theta_R \neq \theta_I$ ) and stationary-point displacement ( $\alpha \neq 2$ ). In accessible quantum regimes both conditions hold to whatever precision matters; predicted deviations in extreme regimes are structurally distinguishable by which handle fails [1, §V B].

*Remark 7.6* (Higher-dimensional scope). The irreducible carrier treated here is the two-outcome, rank-one projective calculus carried by  $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ . Multi-outcome finite systems require direct sums, tensor products, or matrix-algebra extensions of the same dagger/idempotent/trace structure. The present paper proves the generator case; it does not need a Gleason premise for that generator, and it does not claim that every higher-dimensional operational setting has been classified here.

## APPENDIX A. NOTATION

|                                   |                                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$ :                        | fractional order of the accumulation relation; at $\alpha = 2$ the parent term is $\alpha J = 2J$ .                                               |
| $\theta_\alpha$ :                 | phase bookkeeping label $\alpha\pi/2$ ; at $\alpha = 2$ , $\theta_\alpha = \pi$ is the antipodal branch forced by the bar/dagger product closure. |
| $\Theta$ :                        | relative analyzer/projector angle appearing in $\cos^2(\Theta/2)$ ; it is not the same object as $\theta_\alpha$ .                                |
| $k(\tau)$ :                       | memory kernel; $\delta$ -kernel is the Markovian limit.                                                                                           |
| $J^-(x)$ :                        | past light cone of the event $x$ .                                                                                                                |
| $\mathbb{C} \otimes \mathbb{H}$ : | biquaternions, $\cong M_2(\mathbb{C})$ ; central imaginary $i$ , quaternion units $\mathbf{i}, \mathbf{j}, \mathbf{k}$ .                          |
| $\bar{q}, q^\dagger$ :            | quaternionic (bar) and dagger conjugations; $\bar{i} = +i$ , $i^\dagger = -i$ .                                                                   |
| $N(q)$ :                          | reduced norm $q\bar{q} = \det$ ; indefinite metric channel.                                                                                       |
| $\langle qq^\dagger \rangle_0$ :  | Hermitian scalar self-product; positive Born channel.                                                                                             |
| $\mathcal{H}, \mathcal{A}$ :      | Hermitian and anti-Hermitian projections, $\frac{1}{2}(q \pm q^\dagger)$ .                                                                        |
| $Z_\mu$ :                         | holomorphic two-channel source current feeding channel $\mu$ .                                                                                    |
| $\mathcal{J}(\mathbf{Z})$ :       | embedded biquaternionic source-current velocity, not a state vector.                                                                              |
| $E_{\mathbf{n}}$ :                | Hermitian split unit $i(n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k})$ with $E_{\mathbf{n}}^2 = 1$ .                                             |
| $J_\pm$ :                         | closed null source currents $1 \pm E_{\mathbf{n}}$ satisfying $J_\pm^2 = 2J_\pm$ and $N(J_\pm) = 0$ .                                             |
| $P_\pm$ :                         | projectors $J_\pm/2$ .                                                                                                                            |
| $\tau_{\mathcal{B}}$ :            | internal trace readout $2 \operatorname{Re} \operatorname{Sc}(\cdot)$ .                                                                           |
| $R$ :                             | normalized biquaternionic rotor acting by $X \mapsto RXR^\dagger$ .                                                                               |
| $P_0$ :                           | fixed reference idempotent used in the rotor derivation.                                                                                          |
| $W$ :                             | central scalar transition amplitude used as shorthand for the trace-product closure.                                                              |

|                                 |                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------|
| $G, S$ :                        | geometric (dagger-even) and spectral (dagger-odd) channels of $W$ .                                            |
| $h$ :                           | split-complex unit, $h^2 = +1$ .                                                                               |
| $\tau$ :                        | proper-time separation between $x$ and $x'$ ; the argument of the memory kernel. Not the coordinate time $t$ . |
| $\lambda, \hbar_{\text{eff}}$ : | channel width and emergent action scale.                                                                       |

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